

Problem-Solution Fit

Date	15 July 2026
Team ID	SWTID-2026-1616
Project Name	Credit card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Individuals applying for a credit card • Banks and financial institutions • Loan/Credit officers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited financial knowledge • Lack of credit history • Time constraints • Internet/device availability	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Verification ✓ Human judgment ✗ Slow and error-prone Traditional Rule-Based Systems ✓ Simple to implement ✗ Less accurate and inflexible
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Manual approval process is slow (High) • Inconsistent decision-making (Medium) • Long waiting time for applicants (High)	9. PROBLEM ROOT / CAUSE [RC] • Large number of applications • Time-consuming manual verification • Human bias and inconsistency • Difficulty analyzing multiple applicant factors simultaneously	7. BEHAVIOR + ITS INTENSITY [BE] • Frequently checks eligibility before applying (High) • Wants quick decisions (High) • Compares different banks (Medium)
3. TRIGGERS TO ACT [TR] • Customer applies for a credit card • Need for instant eligibility	10. YOUR SOLUTION [SL] Develop a Machine Learning-based Credit Card Approval Prediction	8. CHANNELS OF BEHAVIOR [CH] ONLINE Bank websites, Mobile apps, Credit card portals

check

- Bank wants to automate approvals

4. EMOTIONS

BEFORE / AFTER [EM]

Before: Anxious, uncertain, hopeful

After: Happy if approved; disappointed but informed if rejected

System using **Random Forest** and **Flask** to analyze applicant details and predict approval/rejection instantly. The system provides faster, consistent, and data-driven decisions through an easy-to-use web interface.

OFFLINE

Bank branches, Customer service representatives